3(a)	change in displacement / time (taken)	**B1**
3(b)	by calculation: $v^2 = 42^2 + 23^2 - (2 \times 42 \times 23 \times \cos 54°)$ **or** $v^2 = (42 - 23 \cos 54°)^2 + (23 \sin 54°)^2$ **or** $v^2 = (42 - 23 \sin 36°)^2 + (23 \cos 36°)^2$	**C1**
	$v = 34\ \text{m s}^{-1}$	**A1**
	or	
	by scale diagram: triangle of vector velocities drawn	**(C1)**
	$v = 34\ \text{m s}^{-1}$ (*allow* $\pm 1\ \text{m s}^{-1}$ *if scale diagram used*)	**(A1)**
3(c)(i)	$(\Delta)E = mg(\Delta)h$ **or** $(\Delta)E = W(\Delta)h$	**C1**
	$h = 6100/46$ (= 133 m)	**C1**
	$\theta = \sin^{-1}(133/280)$ $= 28°$	**A1**
3(c)(ii)	force = 6100 / 280 **or** 46 sin 28°	**C1**
	= 22 N	**A1**
3(d)	$v_{(s)} = 280 / 14$ (= $20\ \text{m s}^{-1}$)	**C1**
	$f_o = f_s v / (v - v_s)$ $f_s = 450 \times (340 - 20) / 340$	**C1**
	= 420 Hz	**A1**